

Université Internationale de Rabat

THE INNOVATIVE UNIVERSITY

Hands-on on Quantum Computing

Quantum Gates

2026-2027

Lab's objective

This lab introduces basic Quantum Gates. They are the building blocks of quantum circuits, just as classical logic gates are for conventional digital circuits. Due to the normalization condition, every gate operation U must be unitary, satisfying $U^\dagger U = I$, where $U^\dagger$ denotes the conjugate transpose of U and I is the identity matrix. The number of qubits in the input and output of a quantum gate must be equal; a gate acting on n qubits is therefore represented by a $2^n \times 2^n$ unitary matrix. Unlike many classical logic gates, quantum gates are reversible.

I Single-Qubit Gates

The goal of this part is to :

- Study single-qubit unitary operations.
- Analyze quantum state evolution under basic gates.
- Interpret single-qubit gates as rotations on the Bloch sphere.

It is worth noting that, as stated in Lab1, a single qubit is represented as a normalized vector in a two-dimensional complex Hilbert space : $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $\alpha, \beta \in \mathbb{C}$ satisfy the normalization condition : $|\alpha|^2 + |\beta|^2 = 1$. In vector form :

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Pauli-X (Bit Flip)

It is a Quantum NOT gate :

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

With action on computational basis :

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$

As depicted in Fig. 1,

- Before applying X : the state is at the north pole of the Bloch sphere ($|0\rangle$).
- After applying X : the state moves to the south pole ($|1\rangle$).
- The Pauli-X gate corresponds to a rotation of π radians (180°) around the X-axis.

— Upon measurement, there is a 100% probability of obtaining 1.

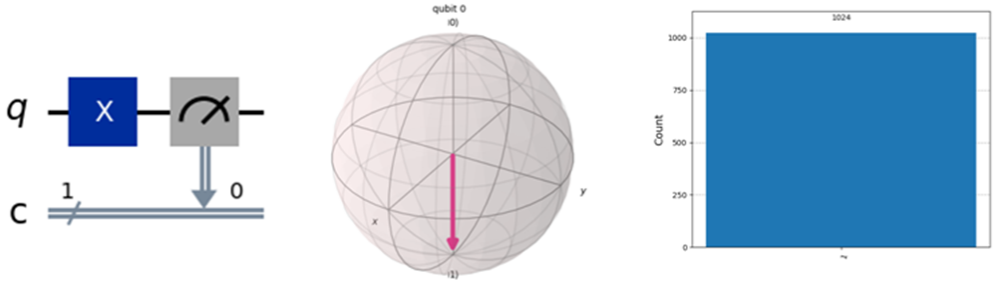


FIGURE 1 – Pauli-X

Pauli-Y

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Applying Y to the computational basis states :

$$Y|0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle,$$

$$Y|1\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -i \\ 0 \end{pmatrix} = -i|0\rangle.$$

As depicted in Fig. 2 :

— Before applying Y : the state is $|0\rangle$.
— After applying Y : $Y|0\rangle = i|1\rangle$.
— The Pauli-Y gate is equivalent to the rotation $R_y(\pi)$ up to a global phase.
— Since global phase does not affect measurement, the probabilities satisfy $P(1) = 1, \qquad P(0) = 0.$

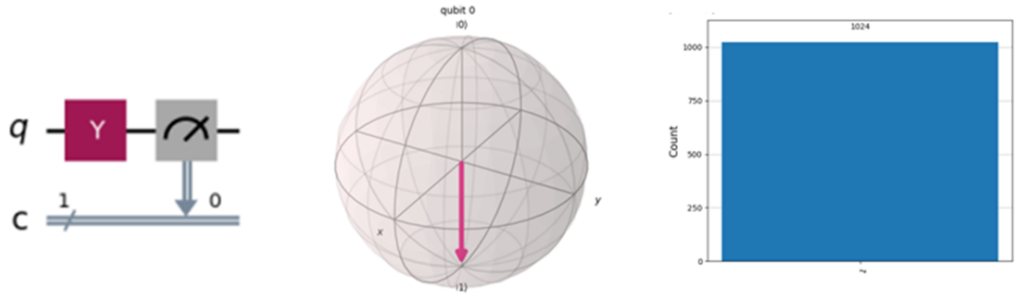


FIGURE 2 – Pauli-Y

Pauli-Z (Phase Flip)

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

As shown in Fig. 3 :

- $Z|0\rangle = |0\rangle$ (north pole remains the same)
- $Z|1\rangle = -|1\rangle$ (south pole acquires a phase -1)
- Global phase does not affect measurement probabilities
- If the initial state is $|0\rangle$, measurement gives 0 with 100% probability

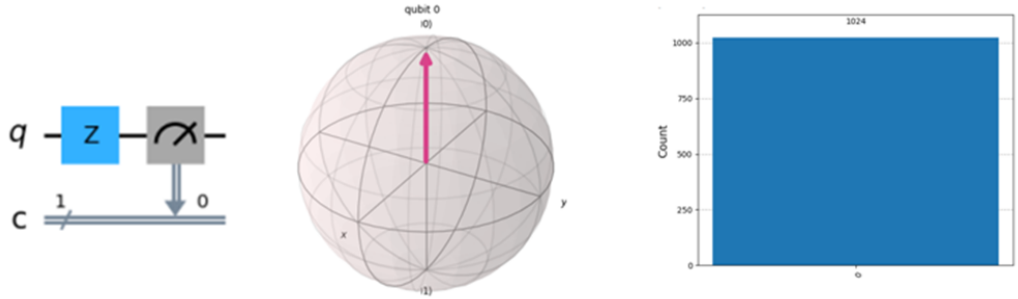


FIGURE 3 – Pauli-Z

Hadamard Gate

The Hadamard gate creates superposition :

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its action on basis states :

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}},$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

As stated in Fig. 4 :

— **Bloch Sphere :** The vector moves from the north pole ($|0\rangle$) to the equator of the Bloch sphere. If you start from $|1\rangle$, it goes to the opposite point on the equator.

— **Measurement :** Since it's a 50/50 superposition, measuring gives approximately 50% $|0\rangle$ and 50% $|1\rangle$ in practice.

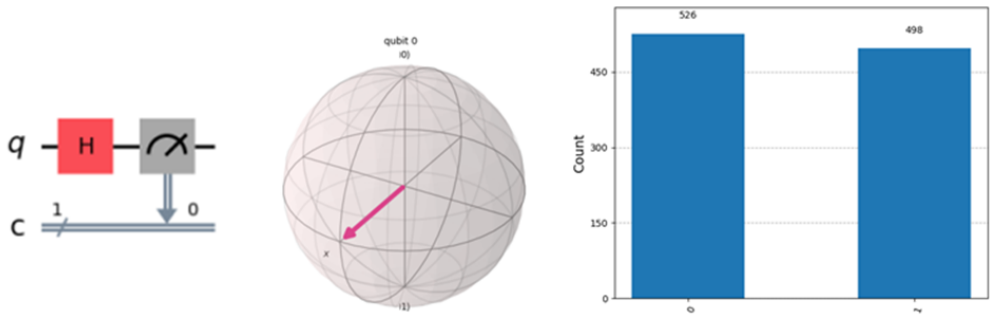


FIGURE 4 – H gate

S Gate

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}.$$

$$S|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle.$$

$$S|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle.$$

As depicted in Fig. 5 :

— $S|0\rangle = |0\rangle, \quad S|1\rangle = i|1\rangle \quad \rightarrow$ adds a phase of $\pi/2$ to $|1\rangle$.
— **Bloch sphere :** For $|0\rangle$ or $|1\rangle$, the vector stays at the poles. For $|+\rangle$, it rotates around the Z-axis by $\phi = \pi/2$.
— **Measurement :** Unchanged if in basis states ; approximately 50% for $|0\rangle$ and $|1\rangle$ if starting from $|+\rangle$.

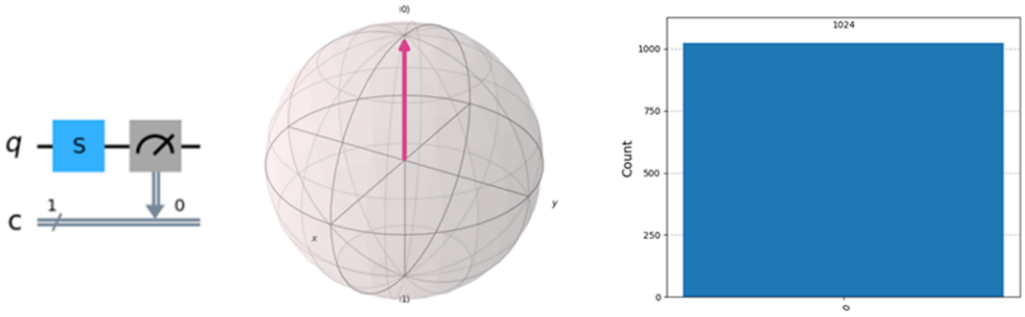


FIGURE 5 – S gate

T Gate

The T gate, also called the $\pi/4$ phase gate, is :

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

Its action on computational basis states :

$$T|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle.$$

$$T|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ e^{i\pi/4} \end{pmatrix} = e^{i\pi/4}|1\rangle.$$

As depicted in Fig. 6 :
— T gate adds a $\pi/4$ phase to $|1\rangle$.
— $|0\rangle$ remains unchanged.
— For superposition states (like $|+\rangle$), the vector rotates around the Z-axis by $\phi = \pi/4$.

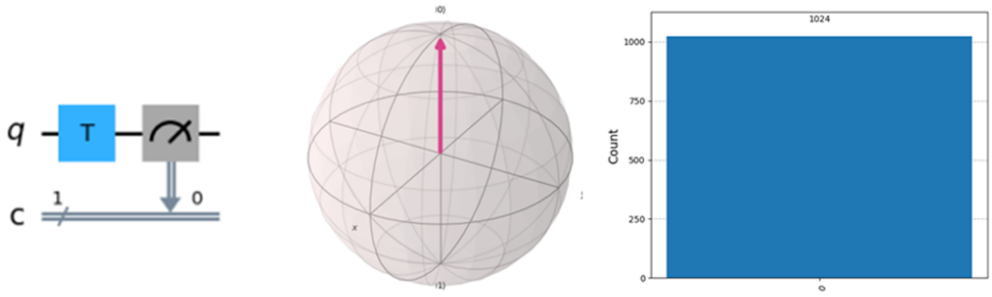


FIGURE 6 – T gate

II Two-Qubit Gates

The goal of this part is to :

- Understand the tensor product structure of multi-qubit systems.
- Study two-qubit unitary operators.
- Analyze entanglement generation.
- Explore controlled quantum operations.

A two-qubit system lives 4 dimensional Hilbert space. The computational basis is :

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix},$$

$$|10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

A general two-qubit state is, with normalization, $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle,$$

Controlled-NOT (CNOT) Gate

The CNOT gate flips the target qubit if the control qubit is $|1\rangle$. Its matrix representation :

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Action on computational basis :

$$\begin{aligned} \text{CNOT}|00\rangle &= |00\rangle, \\ \text{CNOT}|01\rangle &= |01\rangle, \\ \text{CNOT}|10\rangle &= |11\rangle, \\ \text{CNOT}|11\rangle &= |10\rangle. \end{aligned}$$

As depicted in Fig. 7 :

- Hadamard on the control qubit puts it into superposition.
- CNOT entangles the control and target qubits, creating the Bell state $|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$.
- Measurement yields approximately 50% $|00\rangle$ and 50% $|11\rangle$.
- Reduced Bloch spheres show each qubit as completely mixed (vector at the center).

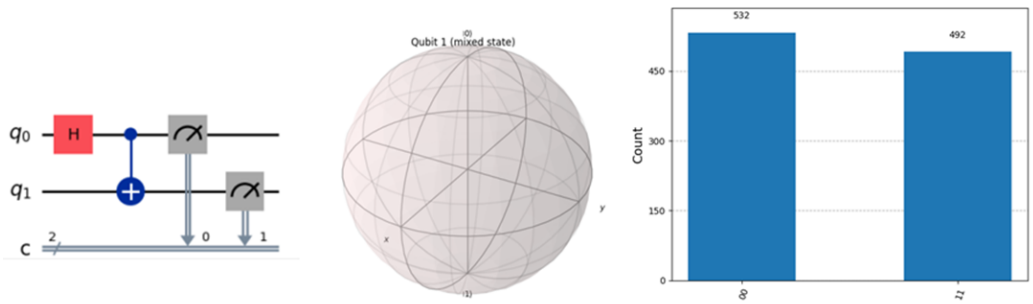


FIGURE 7 – CNOT gate

Controlled-Z (CZ) Gate

Matrix representation :

$$\text{CZ} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

The Controlled-Z gate acts on computational basis states as follows :

$$CZ\,|ab\rangle = \begin{cases} -|11\rangle & \text{if } a = 1 \text{ and } b = 1, \\ |ab\rangle & \text{otherwise.} \end{cases}$$

Equivalently, its action on the standard basis states is :

$$\begin{aligned} CZ\,|00\rangle &= |00\rangle, \\ CZ\,|01\rangle &= |01\rangle, \\ CZ\,|10\rangle &= |10\rangle, \\ CZ\,|11\rangle &= -|11\rangle. \end{aligned}$$

As depicted in Fig. 8 :

— Hadamard on the control qubit (if applied) puts it into superposition.
— Controlled-Z applies a phase flip to the target qubit only when the control qubit is $|1\rangle$.
— The CZ gate creates entanglement between the qubits, producing a Bell-like state

$$|\Phi\rangle = \frac{|00\rangle + |01\rangle + |10\rangle - |11\rangle}{2} \quad \text{(if starting from } |+\rangle|+\rangle\text{)}.$$

— Measurement probabilities depend on the input state ; for $|+\rangle|+\rangle$, each outcome occurs with 25% probability.
— Reduced Bloch spheres show each qubit in a partially mixed state due to entanglement (vector closer to center).

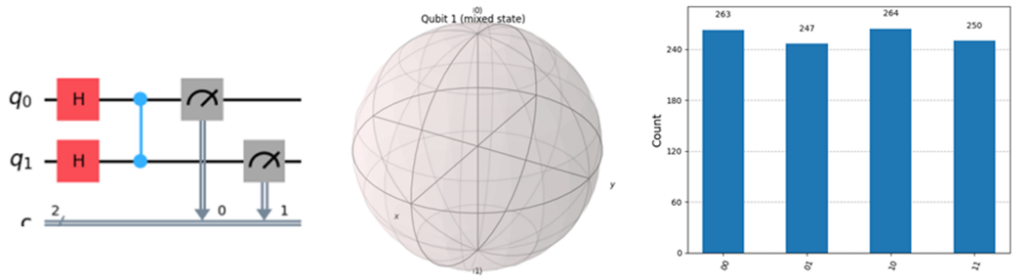


FIGURE 8 – CZ gate

SWAP Gate

The SWAP gate exchanges two qubits :

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Action,

$$\text{SWAP}|01\rangle = |10\rangle,$$
$$\text{SWAP}|10\rangle = |01\rangle.$$

As depicted in Fig. 9 :

— Initial state : $|10\rangle$
— SWAP : $|10\rangle \to |01\rangle$
— No entanglement created
— $P(|01\rangle) \approx 1$
— Both qubits remain pure (Bloch vectors not at origin). **Please note that in Qiskit's convention, qubit 0 is the rightmost bit**.

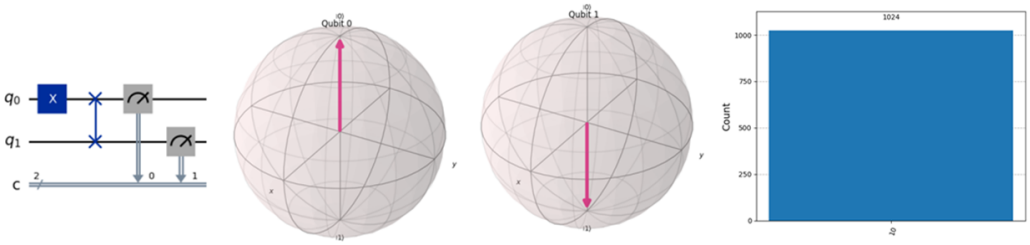


FIGURE 9 – SWAP gate

III Three-Qubit Gates

The goal of this section is to :

— Understand the Hilbert space for three qubits.
— Study multi-controlled gates such as the Toffoli gate.
— Explore universality in quantum computation.

It is worth noting that a three-qubit system lives in a 8-dimensional Hilbert space. Its computational basis is :

$$|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle.$$

A general three-qubit state is, with normalization $\sum_{i=0}^{7} |\alpha_i|^2 = 1$.

$$|\psi\rangle = \alpha_0|000\rangle + \alpha_1|001\rangle + \cdots + \alpha_7|111\rangle,$$

Toffoli Gate (CCNOT)

The Toffoli gate is a **controlled-controlled-NOT** gate :

- The first two qubits are *control qubits*.
- The third qubit is the *target qubit*.
- The target flips if both controls are $|1\rangle$.

Matrix representation (8x8) :

$$\text{Toffoli} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Action on computational basis :

$$\text{Toffoli}|abc\rangle = \begin{cases} |ab\,\bar{c}\rangle & \text{if } a = b = 1, \\ |abc\rangle & \text{otherwise.} \end{cases}$$

Example :

$$\text{Toffoli}|110\rangle = |111\rangle, \quad \text{Toffoli}|101\rangle = |101\rangle.$$

As depicted in Fig. 10 :

- Initial state : $|110\rangle$
- Controls $= 1 \Rightarrow$ target flips
- $|110\rangle \to |111\rangle$
- $P(|111\rangle) \approx 1$
- Pure basis state (no entanglement)

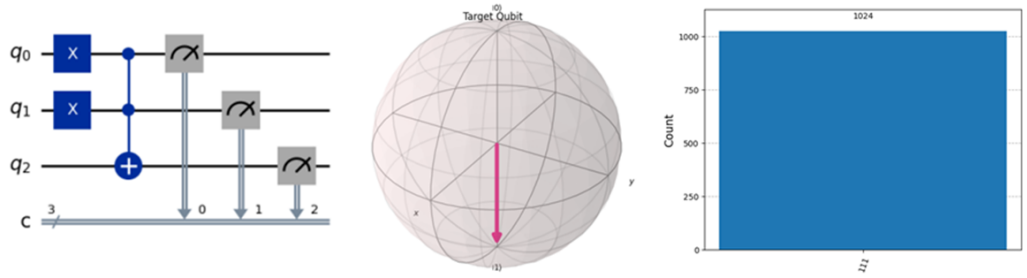


FIGURE 10 – Toffoli gate

1. Fredkin Gate (CSWAP)

The Fredkin gate, also called the **Controlled-SWAP** gate, acts on three qubits :

— Qubit 0 : Control
— Qubit 1 : Target 1
— Qubit 2 : Target 2

The gate swaps the last two qubits if the control qubit is $|1\rangle$, otherwise it leaves the qubits unchanged :

$$\text{CSWAP}|c,a,b\rangle = \begin{cases} |c,b,a\rangle & \text{if } c=1, \\ |c,a,b\rangle & \text{if } c=0. \end{cases}$$

Matrix form (8×8) :

$$\text{CSWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

As depicted in Fig. 11 :

— Initial state : $|110\rangle$
— Controls $= 1 \Rightarrow$ target flips
— $|110\rangle \to |111\rangle$
— $P(|111\rangle) \approx 1$
— Pure basis state (no entanglement)

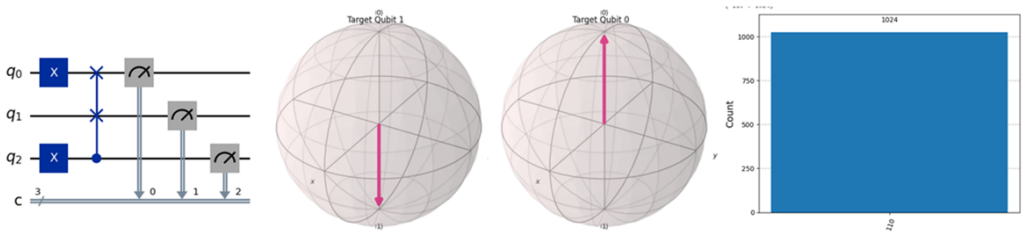


FIGURE 11 – CSWAP gate

2. Controlled-Controlled-U Gate (CC-U)

The **Controlled-Controlled-U** gate generalizes Toffoli.

— **Initial state :** $|110\rangle$
— Both control qubits $= 1 \Rightarrow$ apply unitary U to the target qubit
— **Final state :** $|110\rangle \rightarrow (I \otimes I \otimes U)|110\rangle$
— Measurement depends on the chosen U :
  — If $U = X$, final state $= |111\rangle$ (Toffoli)
  — If $U = Z$, final state $= |11\rangle \otimes Z|0\rangle = -|110\rangle$ (CCZ)
— Entanglement may be created if U produces superpositions

$$\text{CC-}U|a,b,t\rangle = \begin{cases} |a,b\rangle \otimes U|t\rangle & \text{if } a = b = 1, \\ |a,b,t\rangle & \text{otherwise.} \end{cases}$$

As depicted in Fig. 12 :
— Initial state : $|110\rangle$
— Since $c_1 = c_2 = 1$, the controlled-controlled-U gate is activated
— Transformation :

$$|110\rangle \longrightarrow |11\rangle \otimes U|0\rangle$$

— If $U|0\rangle$ is a superposition, the global state may become entangled
— Measurement probabilities depend on the amplitudes of the final state

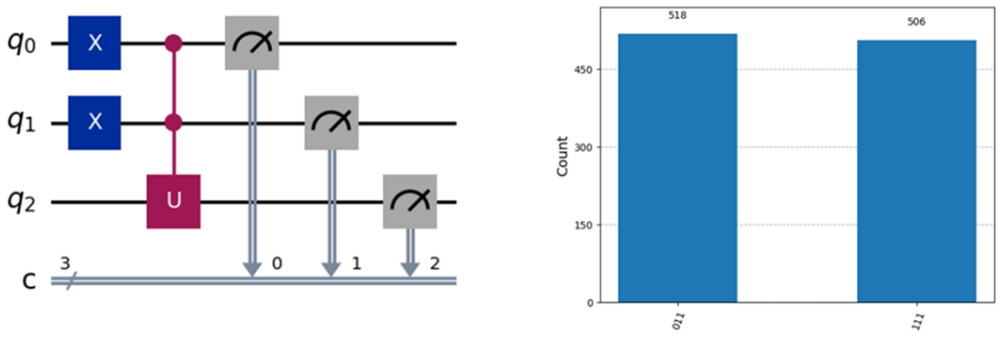


FIGURE 12 – CCU gate